Quantum States and Spectra Lab Report

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1 Introduction

The purpose of this set of experiments was to investigate the quantized nature of atomic energy levels by analyzing the emission and absorption spectra of various gases and dye solutions. When atoms are excited by an electrical discharge, their electrons transition to higher energy states. As these electrons return to lower energy levels, they emit photons of specific wavelengths, which correspond to the energy difference between the two states. As the wavelength of a photon is inversely proportional to it's energy, we can see the energy distribution via the wavelength spectrum.

In this lab, we recorded and analyzed the emission spectra of hydrogen, helium, mercury, and an unknown gas using a PASCO spectrometer. By comparing the measured mercury spectrum to the known mercury spectrum, we were able to calibrate the spectrometer for the other gasses. We did not consider other source of uncertainty aside from the calibration and manufacturers uncertainties, as we made the measurements without difficulty and we had no significant environmental effects. For hydrogen, the observed Balmer series lines were compared with theoretical predictions based on the Rydberg equation, confirming the quantization of energy levels in single-electron atoms. It was shown that the helium atom could not be modeled as a hydrogen atom, as the presence of the second electron adds interaction not considered by the Rydberg equation. The spectrum of an unknown gas was compared with reference data and was determined to be krypton.

In addition to emission spectra, we measured the transmittance and absorbance of dyed liquids to demonstrate how light absorption spectra are related to their perceived colours. The fluorescence of a yellow dyed liquid was also examined to estimate its optical band gap energy and determine whether it exhibited semiconducting properties.

2 Equipment and Methodology

2.1 Equipment

The experimental setup consisted of a PASCO spectrometer connected to a computer running the PASCO Spectrometer software for data collection and wavelength calibration. A discharge tube power supply was used to excite gas discharge tubes containing mercury, hydrogen, helium, and an unknown gas. Both the spectrometer and power supply were turned off when not in use and never turned on for more than 10 consecutive minutes for safety purposes and to prevent wear and tear.

2.2 Discharge tubes

The following steps were performed for all discharge tube measurements (mercury, hydrogen, helium, and unknown gas) and were taken from the lab manual[1]. We inserted the gas tube into the unplugged power supply. We placed the probe of the spectrometer in a clamp that was held at constant height throughout all experiments. The probe was then placed roughly 0.5 cm above the centre of the glass tube. The power supply was then turned on. The probe was adjusted to be at the position of highest intensity of light from the gas tube. The spectrometer was placed in "Analyze Light" mode, and the spectrum was recorded. The power supply was unplugged and the tube allowed to cool prior to removal.

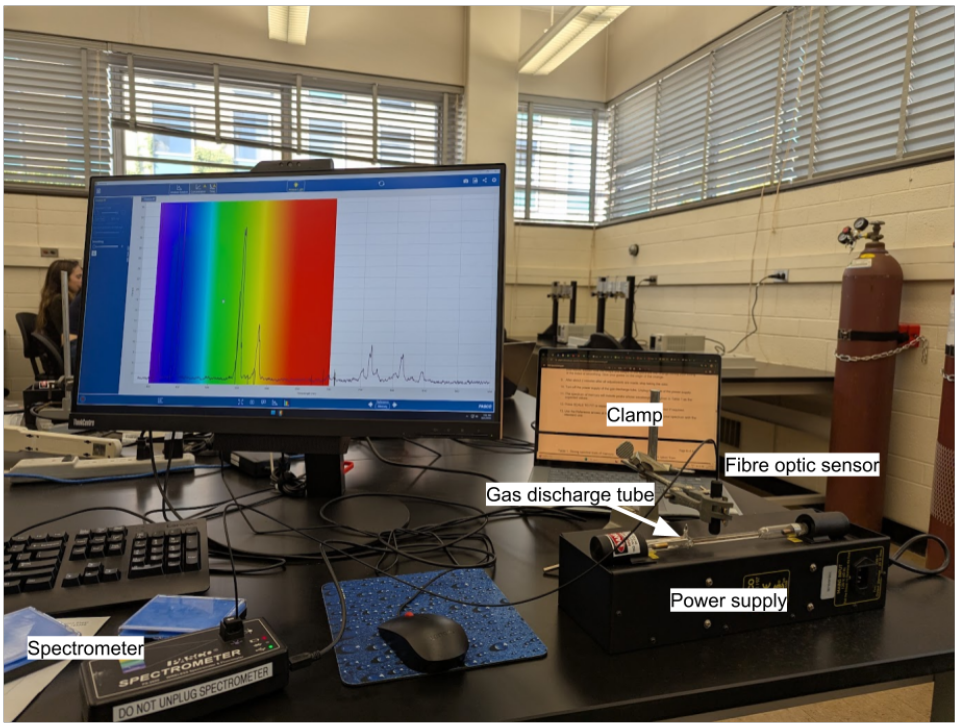

Figure 1: Our labeled lab setup.

2.3 Transmittance and absorbance

The following instructions were also taken from the lab manual [1]. The measurements for this section were made in the "Analyze solution" mode of the spectrometer. The spectrometer was first calibrated dark, and then calibrated with a reference vial of distilled water with the glossy side facing the white light source according to the manufacturers instructions. For the red, green, and blue dyes, the glossy sides were placed facing the white light, and the "Absorbance/Transmittance" option was selected to record the spectra. For the yellow cuvette, the glossy sides were placed facing the blue/purple lights. The spectra were recorded with the "Absorbance/Transmittance", "405-nm excitation", and 500-nm excitation" settings.

3 Results and discussion

All graphical results not seen in the paper can be viewed in appendix B. Sample calculations are available in appendix C.

3.1 Calibration (mercury)

Our first step was to calibrate the spectrometer by comparing the spectra of a known substance (mercury) to its theoretical spectra. By plotting the wavelengths of our measured peaks against the known wavelengths of the peaks of mercury, we can determine a relation between them and use that relation to estimate the true wavelengths measured for other gases. We were unable to detect the shortest-wavelength peak, however this may have been because the peak was out of range of the spectrometer; according to PASCO, the spectrometer can only detect wavelengths greater than 380 nm[2]. Additionally, shorter wavelengths result from higher-energy transitions, which require more energy to excite the atoms into higher-energy states. Since the amount of voltage (and thus energy) supplied to the gas is limited, few atoms are able to reach the necessary excited states. The uncertainty for the wavelength measurements was taken as the resolution stated by the manufacturer of 3 nm [2].

Colour	Violet	Violet	Blue	Green	Yellow	Yellow
Expected λ (nm)	404.6565	407.7837	435.8328	546.0735	576.9598	579.0663
Expected Energy (eV), eV	3.0643	3.0408	2.8451	2.2708	2.1492	2.1414
Experimental λ ($\pm$3 nm)	N/A	414	445	548	581	583
Experimental Energy ($\pm$ 0.0x eV)	N/A	2.99 (x=2)	2.79 (x=2)	2.26 (x=1)	2.14 (x=1)	2.13 (x=1)

Table 1: Strong spectral lines of mercury[1]. Energy in eV was calculated as $E = \frac{1240}{\lambda}$[1]

3.1.1 The Calibration Curve

After processing our data, we found it matched a linear fit with the equation $y = (1.03 \pm 0.02)x - (20 \pm 10)$, where y is the true wavelength of the mercury peaks and x is our measured peak wavelength. Due to the strong fit, with a χ^2 value of 0.028 and residuals centered evenly around the origin, we believe this calibration is accurate and will use it to calibrate our other wavelength data for this lab.

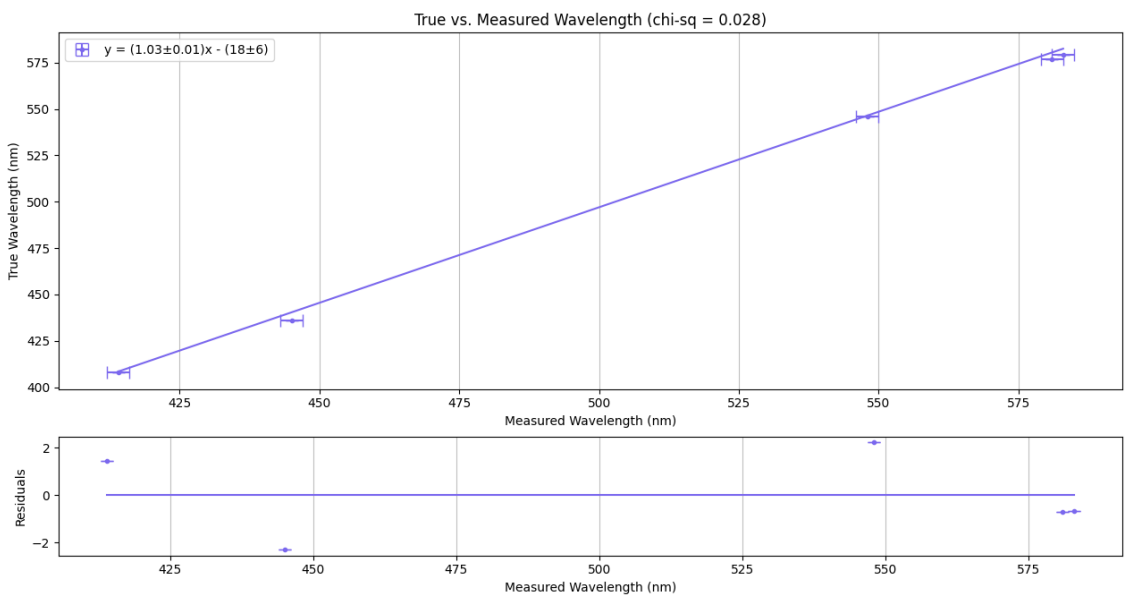

Figure 2: The calibration line for our spectrometer, displayed with relevant goodness of fit metrics (residuals and chi squared). The error in x is $\pm$3 nm, while the error in y is $\pm$0.0001 nm. The variance of y is 4.4874, so the standard deviation is 2.118 nm, which is less than our reading uncertainty of 3 nm. Thus, we take our uncertainty in y as 3 nm (see Section 3.7).

3.2 Quantum States of Hydrogen

3.2.1 Expected Energies and Wavelengths

The expected energy levels for the Hydrogen atom can be found using the equation:

$$\frac{hc}{\lambda} = R_{EH} Z^2 \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$

[1] Where $R_{EH} = 13.605693$ eV is the Rydberg constant for Hydrogen, n is the quantum number of the excited state of the electron, m is the quantum number of the ground state of the electron, and Z is the

number of protons in the nucleus, which is just 1 for Hydrogen. Since $E = hf = \frac{hc}{\lambda}$, rearranging to isolate for λ yields the final equation:

$$\frac{1}{\lambda} = R_H(\frac{1}{m^2} - \frac{1}{n^2}) \tag{1}$$

Where $R_H = 10973731.57$ m^{-1} is the Rydberg constant for inverse wavelength. Similarly, we can calculate the energy of any state of hydrogen with quantum number n with the equation:

$$E = \frac{R_{EH}}{n^2} \tag{2}$$

We will use these equations to calculate the values present in Table 2. Note that the spectral lines in this table are those of the Balmer series (n = 2).

Quantum number	3	4	5
Expected energy of quantum state (eV)	-1.51	-0.85	-0.544
Expected energy of spectral line (eV)	1.89	2.55	2.86
Expected wavelength of spectral line (nm)	656	486	434
Experimental wavelength of spectral line (uncalibrated) (±3 nm)	658	493	N/A
Experimental wavelength of spectral line (calibrated) (±3 nm)	660	489	N/A

Table 2: The quantum numbers, expected energies and wavelengths, and experimentally measured wavelengths for the Balmer series of Hydrogen with m = 3, 4, 5.

The experimental values agree with the theoretical wavelengths within the uncertainties, however we were once again unable to detect the shortest-wavelength peak. As mentioned previously in subsection 4.1, this is due to limitations in our experimental setup, due to our spectrometer's limited detection range and the limited amount of energy being supplied to the gas, making it difficult for the atoms to be excited to the necessary high-energy states.

The relative intensities of the lines will vary for the same atoms, as the gas atoms themselves are concentrated randomly throughout the tube, their electrons will be excited randomly, and they will emit radiation at random times in random directions. Taking these factors into consideration, it is sensible that the relative intensities detected from the same discharge tube will still vary from experiment to experiment.

3.3 Quantum States of Helium

Electron configuration of initial state	Electron configuration of final state	Ref. intensity	Ref. wavelength λ_r (nm)	Measured wavelength λ_{exp} (nm)	E_n (eV)	E_m (eV)	Selection rules		
							Δn	Δl	ΔJ
1s2p (1)	1s^2 (0)	1000	58.433	N/A	-3.34	-24.57	-1	-1	-1
1s3s (1)	1s2p (2)	200	706.519	710 ± 3	-1.84	-3.60	-1	1	1
1s3p (1)	1s2s (1)	500	388.865	394 ±3	-1.56	-4.75	-1	-1	0
1s3d (3)	1s2p (2)	500	587.562	591 ± 3	-1.49	-3.60	-1	-1	-1
1s3d (2)	1s2p (1)	100	667.815	671 ± 3	-1.49	-3.35	-1	-1	-1
1s3p (1)	1s2s (0)	100	501.568	504 ± 3	-1.48	-3.95	-1	-1	-1
1s4d (1)	1s2p (2)	200	447.148	451 ± 3	-0.82	-3.60	-2	-1	1

Table 3: Summary of electron transitions and selection rules.

We could not obtain a reading for the 58.433 nm reference wavelength as the PASCO spectrometer only had a certified range of 380-950 nm[2]. Almost all of the wavelengths we were able to measure with the spectrometer agreed with the given values of wavelength within the uncertainty, with one (the 1s3p (1) to 1s2s (1) transition) differing by less than 1 nm, which is well within our stated uncertainty.

In the helium emission spectrum, three distinct spectral lines correspond to the transitions between the n=3 and n=2 levels. These arise because helium's energy levels are split into 3 distinct sublevels (3s, 3p,

and 3d)[3]. We could not observe the same result for hydrogen as there is only one electron, meaning that all energy levels at n=3 would be degenerate so there would only be one observable line. Equation (2a) is not applicable to the helium atom because atomic helium has two electrons that exert forces on each other, compared to hydrogen only having one electron. This means that the potential energy of the helium electrons have dependencies on more than just energy levels and atomic number, compared to the Coulomb-type relation of the hydrogen electron.

The selection rules in terms of all possible Δ n, Δl and ΔJ for the relevant experimental transitions are as follows [3] :

1. $\Delta n =$ any integer (although in the experiment the values are all negative as they are emission spectra)
2. $\Delta l = \pm 1$
3. $\Delta J = 0, \pm 1$ but $0 \rightarrow 0$ is not allowed

The values in the table follow the rules, as the only time $\Delta J = 0$ is a $1 \rightarrow 1$ transition.

3.4 Determining the Unknown Gas

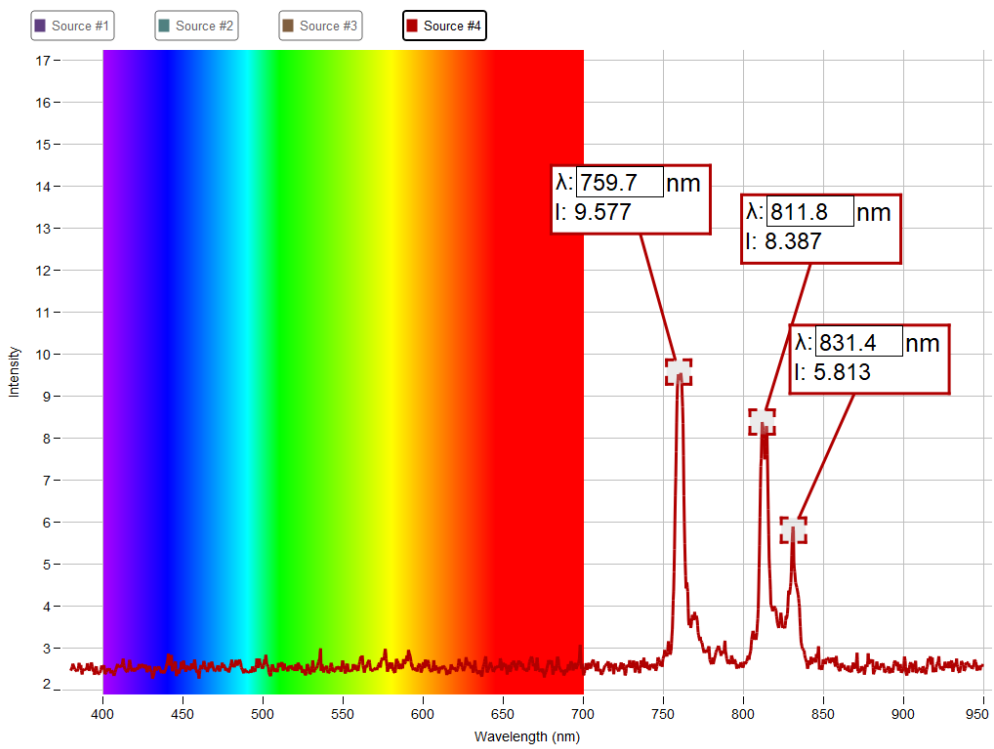

Figure 3: The spectrum of the unknown gas. A reference spectrum is not shown as the reference for our best guess (Krypton) was not present in the software.

Gas	1	2	3	4	5	6	7	8	9
Neon	540	585	622	640	660	693	703	717	725
Argon	427	435	440	459	473	476	502	676	
Krypton	427	432	557	587	646	759	769	810	826
Xenon	484	529	534	542	597	605	610	681	699
Sodium	588.995	589.592							
Nitrogen	404	444	463	501	568	575	594	648	661
Oxygen	412	423	533	559	605	616	626	637	645

Table 4: A table containing information on gases and their known peak wavelengths. [1]

Experimental Wavelength (λ_r) ($\pm$3 nm)	760	812	831
Calibrated Experimental Wavelength (λ) ($\pm$3 nm)	764	818	838
Intensity ($\pm$0.001)	9.577	8.387	5.813
Relative Intensity ($\pm$0.0001)	1.0000	0.8757	0.6070

Table 5: Wavelengths and intensities of spectral lines of the unknown gas. Note that we were only able to observe longer-wavelength peaks, arising from lower-energy transitions.

From the comparison between the three wavelengths we were able to measure and the wavelengths of known gases given in Table 4, we believe the unknown gas is Krypton. Our calibrated experimental wavelengths are closest to the wavelengths seen in Krypton's spectrum within our uncertainty.

3.5 Transmittance and Absorbance

The calculated absorbances were calculated with the provided formula[1] .

$$A = -\log_{10} T$$

All the calculated wavelength uncertainties are reported as $\pm$3nm, as that is the resolution stated by the manufacturer. Uncertainty for the absorbances was calculated as

$$\Delta a = 0.434 \times \frac{\Delta\lambda}{\lambda}$$

. The largest value for this was less than one count of the least significant digit, so we report uncertainties of all absorbances as $\pm$0.001. We do not believe that there is reason to have greater uncertainty, as we followed steps to manufacturer specifications and we had no difficulty in taking measurements.

3.5.1 Blue

Wavelength (nm)	380	392	484	602	656	730
Experimental Transmittance	47.355	24.997	57.576	2.717	33.315	97.637
Calculated absorbance	0.325	0.602	0.240	1.566	0.477	0.010
Experimental Absorbance	0.323	0.602	0.240	1.566	0.477	0.010

Table 6: Experimental and calculated absorbance tables for wavelengths of notes for the blue dyed liquid.

3.5.2 Red

Wavelength (nm)	380	434	535	592	730
Experimental Transmittance	37.398	2.074	5.598	33.051	97.799
Calculated absorbance	0.427	1.683	1.252	0.481	0.010
Experimental Absorbance	0.422	1.702	1.254	0.481	0.010

Table 7: Experimental and calculated absorbance tables for wavelengths of notes for the red dyed liquid.

3.5.3 Green

Wavelength (nm)	378	452	535	607	730	910	945
Experimental Transmittance	2.391	0.1	8.922	2.429	43.343	46.825	39.368
Calculated absorbance	1.621	3.000	1.050	1.615	0.363	0.330	0.405
Experimental Absorbance	1.250	3.000	1.050	1.629	0.363	0.330	0.405

Table 8: Experimental and calculated absorbance tables for wavelengths of notes for the green dyed liquid.

3.5.4 Discussion

Within the visible light spectrum (400-700nm), the transmittance is highest at the wavelengths near the colour of the dye itself. This can be seen on the tables or more clearly in the plots in the appendix. The absorbance is higher at the other wavelengths in the visible light spectrum. For example, the green dyed liquid has a peak transmittance at 535.2 nm, which is the wavelength of green photons. It has low transmittance/high absorbance on both sides of the visible light spectrum (there is a higher transmittance value in the near infrared range but this can be ignored for the discussion purposes as it does not affect the colour we see). Thus, in white light, we will perceive it as green, as more green photons pass through to our eyes and relatively fewer red and blue (and other visible wavelength) photons are transmitted. The same principle applies to the other colours.

3.6 Fluorescence

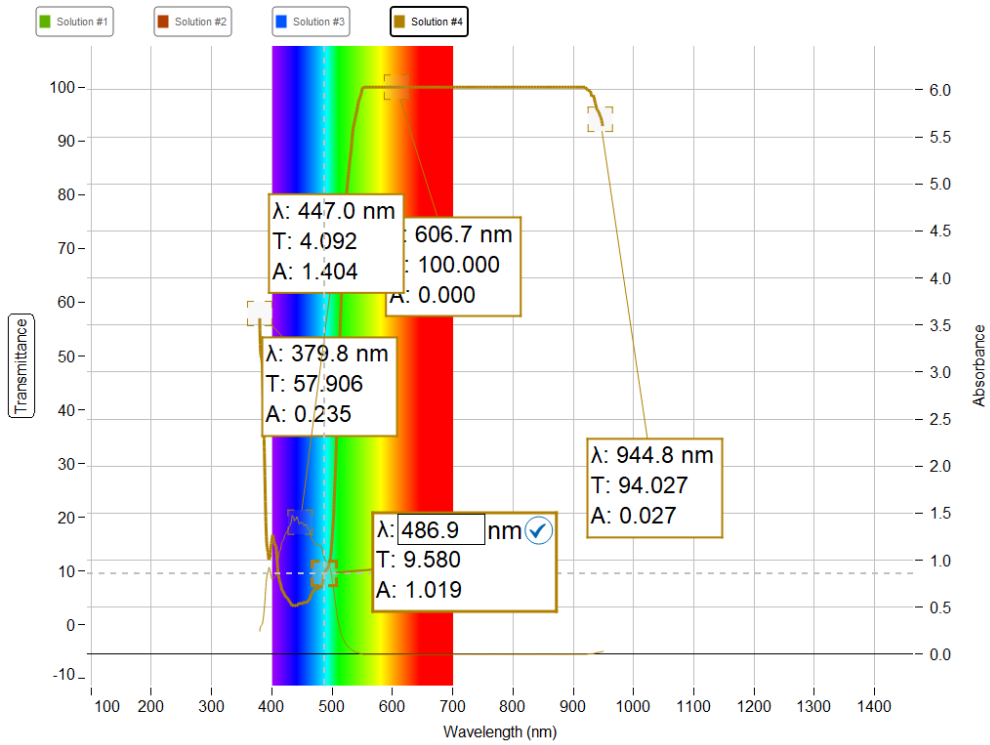

Figure 4: The Absorbance/Transmittance curve for the yellow dyed liquid. The Absorption edge can be seen at 487 ± 3 nm, where the absorbance drops the most (i.e. the rate of change is at it's maximum.)

3.6.1 Discussion

Using the provided band gap equation, we can find the band gap of the yellow dye.

$$E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$$

$$E = \frac{1240}{487} = 2.54 \pm 0.02 \text{ eV}$$

Given that this band gap value falls within the 0-3 eV band gap characteristic of semiconductors [4], then we can say that the yellow dyed liquid demonstrates semiconducting properties.

3.7 Uncertainty Propagation

The following are the formulae we used to propagate uncertainties for experimental values in this paper. For each formula, we list the tables, figures and other calculations (listed via their subsection number) that make use of the formula. We took the reading error in y to be 3 nm, as the spectrometer manual mentioned a resolution of 3 nm. Sample calculations are provided in Appendix B.

3.7.1 Wavelength to Energy

$$\Delta E = E\frac{\Delta\lambda}{\lambda}$$

This formula was used for Tables 1 and the band gap energy calculation (section 3.6.2).

3.7.2 Uncertainty in Calibrated Wavelength

$$\Delta y = \max(\sqrt{\frac{\Sigma[y_i - (mx_i + b)]^2}{N-2}}, 3)$$

This formula was used for Figure 2 and Tables 2, 3, and 5, as well as our uncertainty calculations for the uncertainties in the slope and y-intercept of our calibration curve (note we use the value in the left side of the max function).

3.7.3 Uncertainty in Slope and Y-Intercept

$$\Delta m = \Delta y \sqrt{\frac{N}{N\Sigma x_i^2 - (\Sigma x_i)^2}}$$

$$\Delta b = \Delta y \sqrt{\frac{\Sigma x_i^2}{N\Sigma x_i^2 - (\Sigma x_i)^2}}$$

These formulas were used for Figure 2.

4 Appendices

4.1 Appendix A: References

References

[1] University of Toronto Department of Physics. Spectrometry laboratory manual. `https://www.physics.utoronto.ca/~phy224_324/LabManuals/OLD_Spectrometry.pdf`, n.d. Accessed October 7, 2025.

[2] Vernier Software & Technology. Determining the rydberg constant. `https://www.vernier.com/files/sample_labs/VSPEC-08-COMP-rydberg_constant.pdf`, n.d. Accessed October 7, 2025.

[3] A. J. Hewitt. Selection rules notes. `https://jila.colorado.edu/~ajsh/astr5110_13/notes/selec.pdf`, 2013. Accessed October 7, 2025.

[4] W. H. Strehlow and E. L. Cook. Compilation of energy band gaps in elemental and binary compound semiconductors and insulators. *Journal of Physical and Chemical Reference Data*, 2(1):163–200, 1973.

4.2 Appendix B: Images

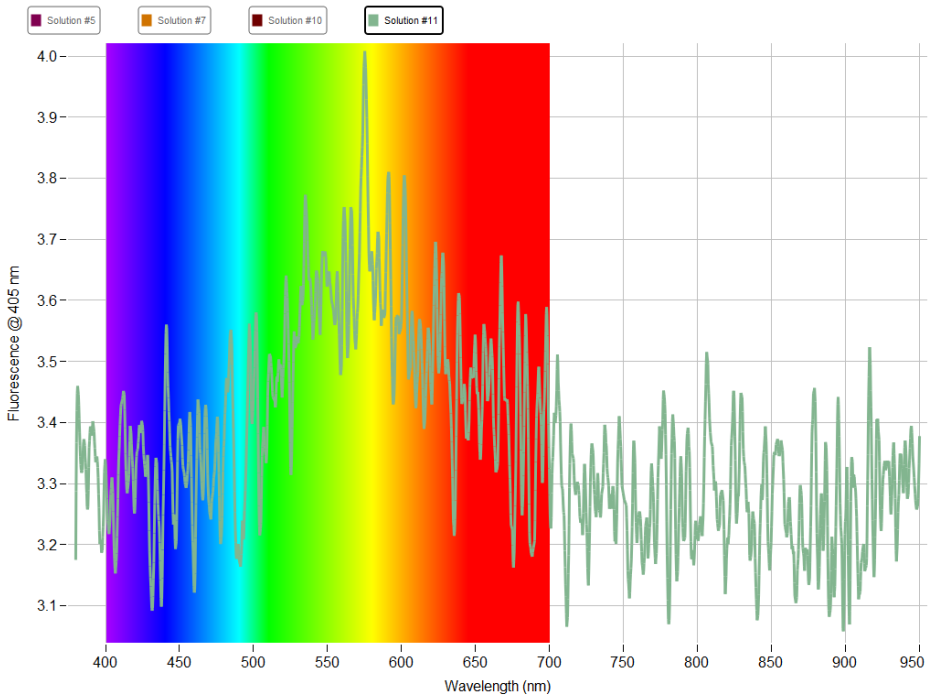

Figure 5: 405 nm excitation spectrum

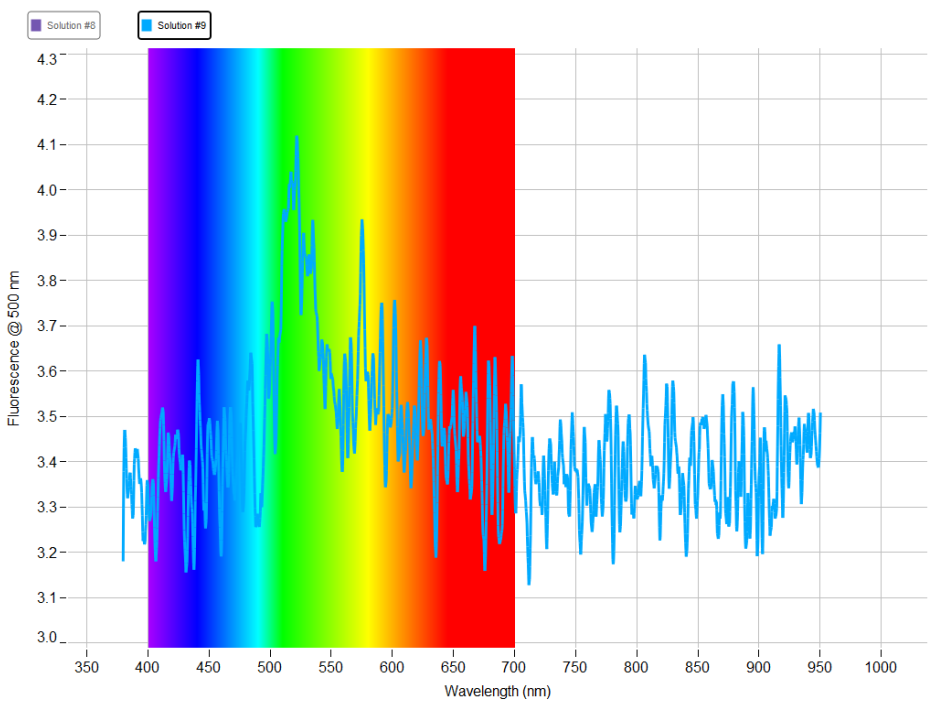

Figure 6: 500 nm excitation spectrum

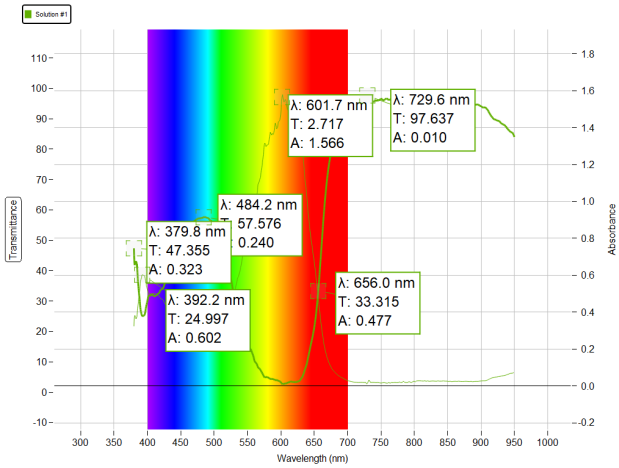

Figure 7: Blue dyed liquid transmittance/absorbance spectrum

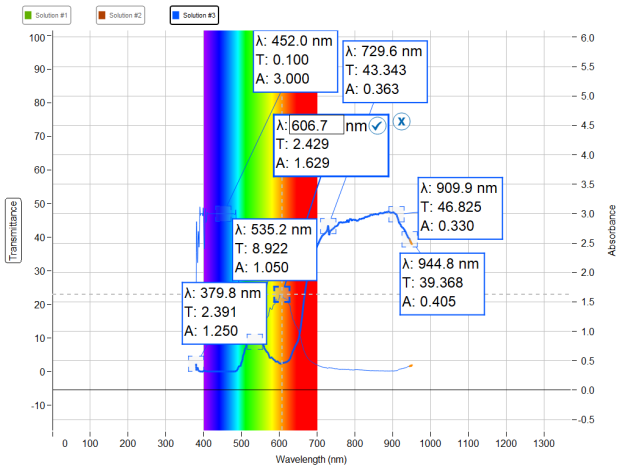

Figure 8: Green dyed liquid transmittance/absorbance spectrum

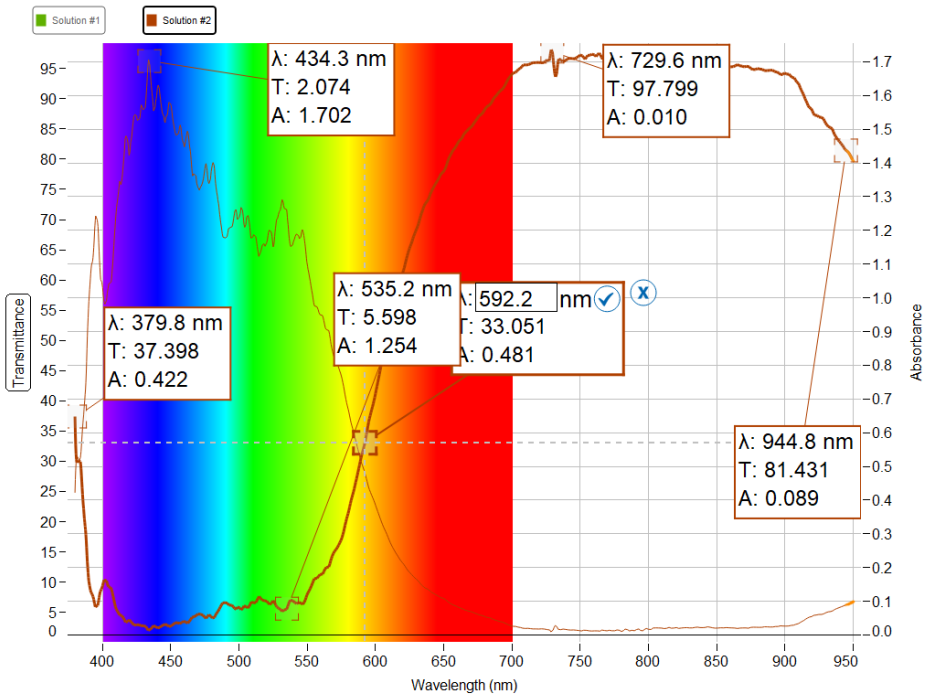

Figure 9: Red dyed liquid transmittance/absorbance spectrum

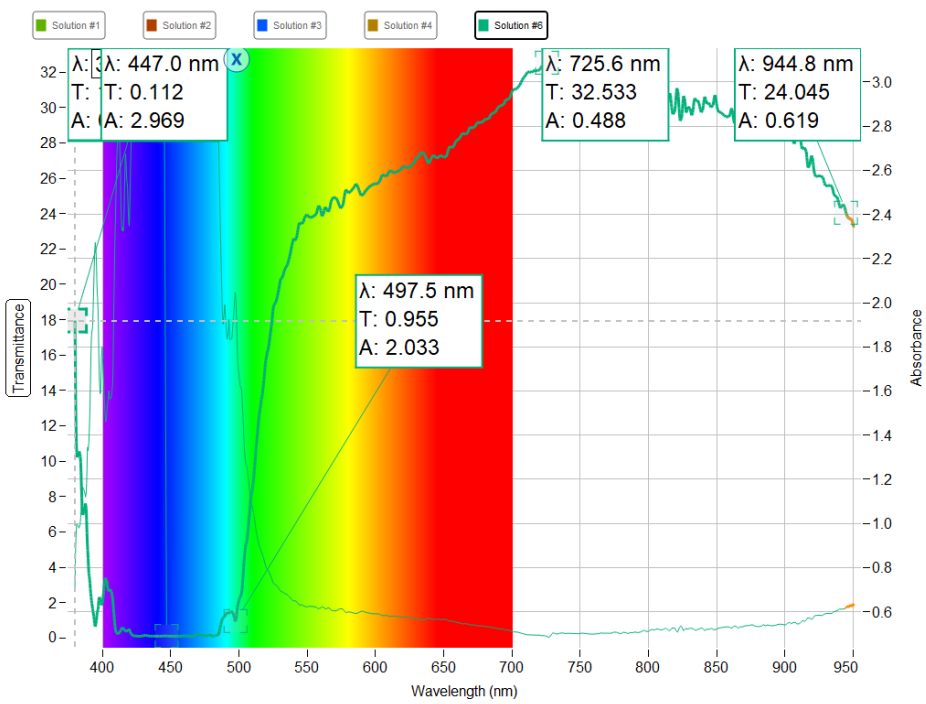

Figure 10: Yellow dyed liquid transmittance/absorbance spectrum

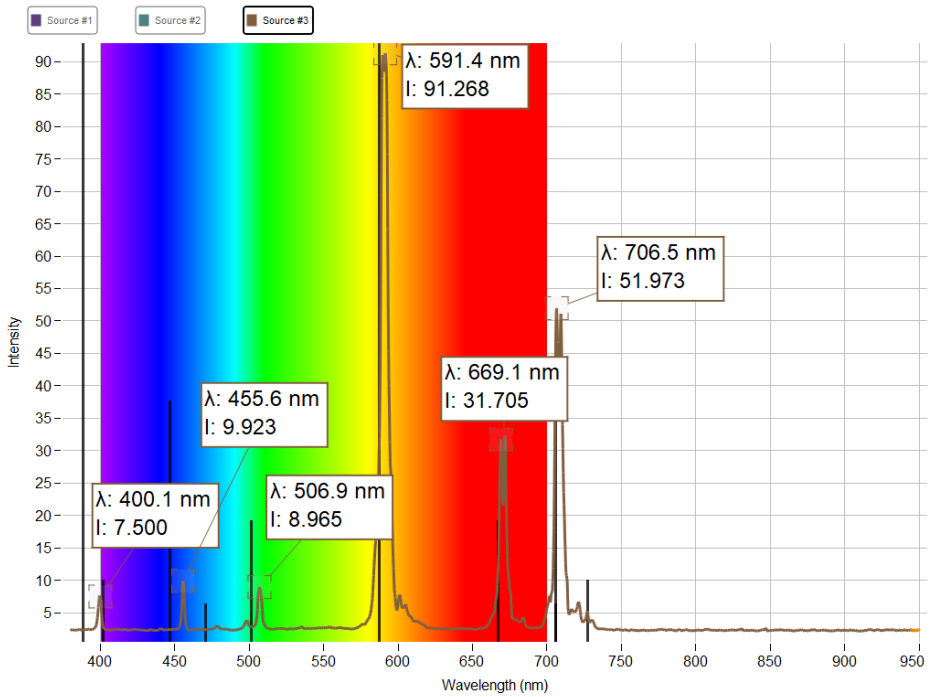

Figure 11: Helium excitation spectrum

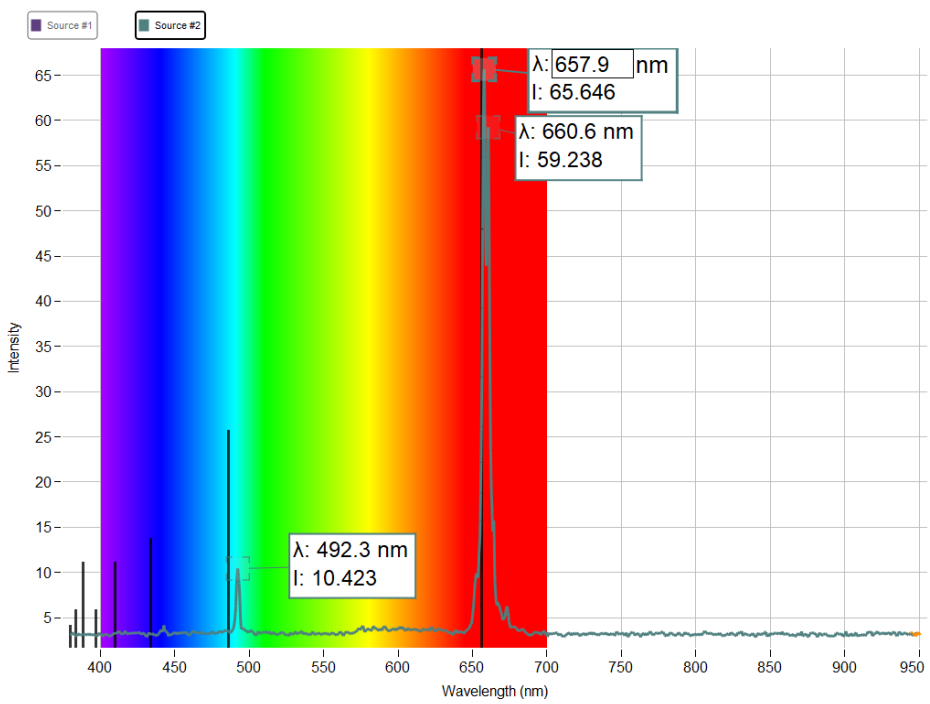

Figure 12: Hydrogen excitation spectrum

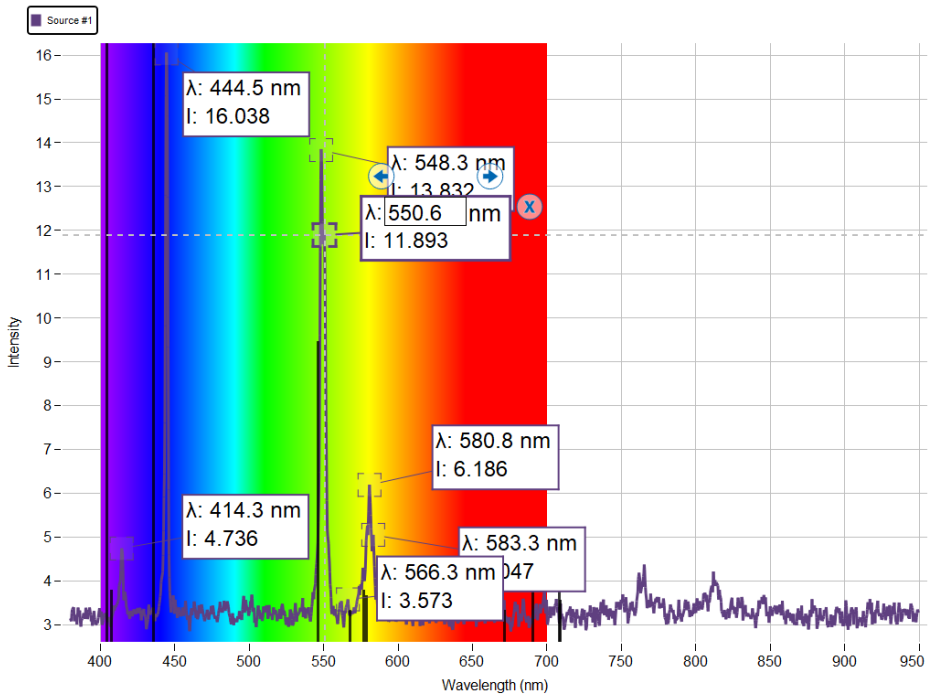

Figure 13: Mercury excitation spectrum

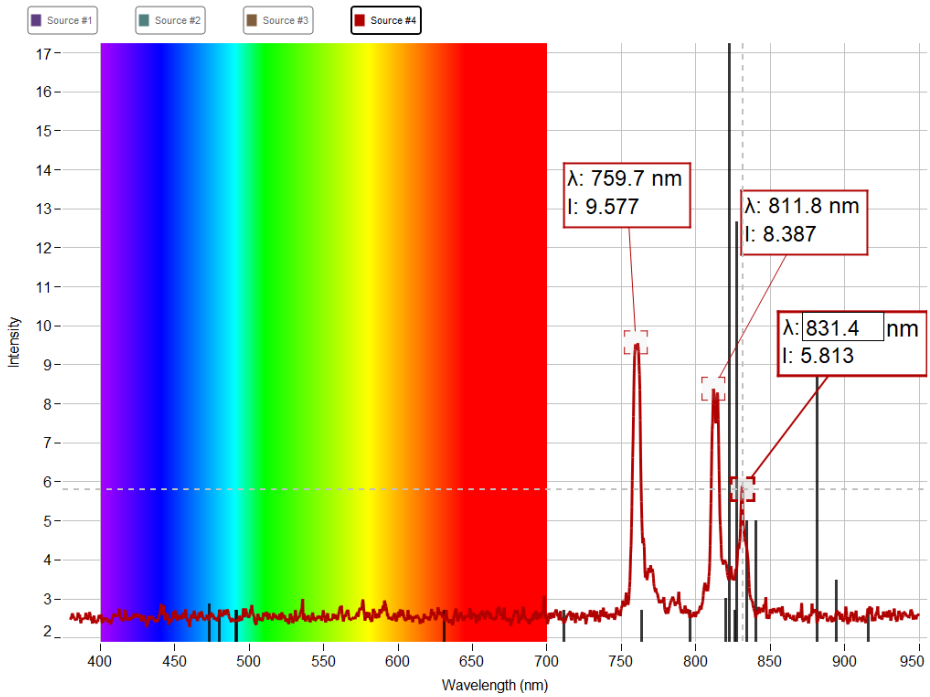

Figure 14: Unknown gas excitation spectrum

4.3 Appendix C: Sample Calculations

4.3.1 Wavelength Propagation

4.3.2 Wavelength to Energy

We perform a sample calculation for the wavelength to energy uncertainty using the mercury calibration data with experimental wavelength 414 ± 3 nm:

$$\Delta E = (2.993)(\frac{3}{414})$$

$$\Delta E = 0.0217 = 0.02$$

4.3.3 Calibrated Wavelength

Here, we perform a sample calculation for the entire mercury spectrum:

$$\Delta y = \max(\sqrt{\frac{(407.7837 - [1.03 \cdot 414 - 18])^2 + \cdots + (579.0663 - [1.03 \cdot 583 - 18])^2}{5-3}}, 3)$$

$$\Delta y = \max(2.118, 3) = 3$$

4.3.4 Calibration Slope and Y-Intercept

Once again, we perform a sample calculation for the entire mercury spectrum:

$$\Delta m = (2.118)\sqrt{\frac{(5)}{(5)(414^2 + 445^2 + \cdots + 583^2) - (441 + 445 + \cdots + 583)^2}}$$

$$\Delta m = 0.01335 = 0.01$$

$$\sqrt{\frac{(414^2 + 445^2 + \cdots + 583^2)}{}}$$